Metabolic Adaptations to Sprint Interval Training: A Literature Review

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Abstract

Objective: To present an overview of the current scientific literature regarding metabolic adaptations and performance improvements resulting from sprint interval training. To examine how those adaptations compare to endurance training adaptations for the purpose of demonstrating a time-efficient approach to improve aerobic capacity and reduce metabolic risk factors.

Methods: The PubMed database was searched utilizing the terms *sprint interval training, metabolic adaptations,* and *endurance training* singularly or in combination. PubMed, Google Scholar, textbooks, and websites were cross-referenced to define terminology determined to be necessary for proper reader understanding.

Conclusion: Sprint interval training, characterized by less than 20 minutes of total workout intensity over a 2-week period, improves muscle oxidative capacity and functional performance as effectively as traditional endurance training. Due to the potential for increased compliance, sprint interval training may demonstrate greater clinical success than endurance training as a preventative health strategy to reduce metabolic risk factors.

Keywords: *sprint interval training, metabolic adaptations, endurance training, aerobic, anaerobic, muscle oxidative capacity, mitochondrial enzymes*

Introduction

Conventional wisdom holds that in order to increase aerobic capacity as evidenced by improved extended duration performance, an individual has to engage in extended duration, steady state exercise referred to as traditional endurance training. Endurance is defined as the ability to sustain a prolonged stressful effort or activity or to remain active for a long period of time with the ability to resist fatigue.[1,2] Traditional endurance training; therefore, includes distance running, cycling, swimming, and other relatively low intensity physical exercises performed for longer than 30 minutes which depend primarily on the use of oxygen to meet the energy demands of training.[3,4] The belief that improving aerobic capacity requires endurance training has been challenged as relatively recent research data has been published revealing that sprint interval training, generally characterized by less than 20 minutes of total intensity over a 2-week period, can double endurance performance.[5-8]

Because of the short duration of maximum effort, sprint interval training would characteristically be described as anaerobic, meaning without oxygen, training. Therefore, anaerobic training is defined as training in the absence of oxidative metabolic pathways in order to trigger anaerobic metabolism. Anaerobic training is most commonly used by non-endurance athletes to build mass, strength, power, and speed. Because anaerobic training is designed to develop muscle energy systems differently than aerobic, endurance training, anaerobic training is characteristically used to improve performance in short duration, high intensity activities comparable to sprint interval training, i.e. sprinting, jumping, weightlifting. As a result, no oxidative metabolic changes or improved endurance performance would typically be expected from a regimen of sprint interval training.

Numerous physiological adaptations result from endurance training which improve an individual's exercise capacity commonly measured by performance and V_{O_2} max. According to traditional thought, increased mitochondrial density improves respiratory control leading to improved substrate level utilization, i.e. increases in maximal activity of mitochondrial enzymes, with changes in the tricarboxylic acid cycle (TCA cycle) and electron transport chain.[6] However, the idea that low-volume, short-duration, high-intensity training has the ability to produce adaptations leading to improved endurance performance that relies on aerobic metabolism is the driving force behind recent research.

In contrast to extended duration endurance training, sprint interval training minimizes lactate accumulation, i.e. enhances muscle buffering capacity,[8-10] and decreases glycogen utilization in addition to improving mitochondrial enzyme activity.[7] Therefore, in order to understand the role of sprint interval training as a viable aerobic conditioning program, it is important to examine the concept of sprint interval training, the various adaptations that it influences, the significance of the data during its performance, and the mechanisms by which it operates. With an established definition of sprint interval training and presentation of adaptations as well as preliminary comparisons to endurance training adaptations, further interpretation of the research data is necessary to adequately establish the aerobic effects of sprint interval training.

Since sprint interval training involves short, intense, all out bursts, expected anaerobic adaptations would include significantly depleted ATP –CrP system and glycolytic components with negligible changes seen in aerobic adaptations. However, the presented data shows significant aerobic adaptations evidenced by increased mitochondrial enzyme activity including the key mitochondrial marker cytochrome c oxidase and the TCA cycle components citrate synthase, malate dehydrogenase, and succinate dehydrogenase. Furthermore, increases in resting muscle glycogen indicate reduced anaerobic activity as muscle glycogen hydrolysis would serve to provide energy for ATP-PCr systems and glycolysis. Since more muscle glycogen is available post-sprint interval training, some energy must have been provided via aerobic processes.[5-8,11-14] More direct evidence of aerobic improvement includes increases in V_{O_2} max and muscle oxidative enzyme activities which directly involve oxygen utilization. Finally, improved time trial performance in endurance activities outwardly demonstrates the physiological adaptations previously discussed.

Methods

The PubMed database was searched utilizing the terms *sprint interval training, metabolic adaptations, and endurance training* singularly or in combination. Searches were limited to clinical trials, reviews, and comparative studies subsequent to January 1992. In addition, searches on Martin Gibala and Kirsten Burgomaster, two prominent researchers on the topic of sprint interval training, were conducted. References for articles determined to be relevant were evaluated and searched to further develop the working material for the purpose of this review. PubMed, Google Scholar, textbooks, and websites were cross-referenced to define terminology determined to be necessary for proper reader understanding.

Discussion

With regard to exercise metabolism, assumed knowledge reduces the need to systematically present each integrated step of the ATP-CrP system, glycolysis, the TCA cycle, and the body's adaptations to prolonged anaerobic and aerobic conditioning programs.

However, a discussion of exercise metabolism necessitates a limited presentation of several key component steps used to provide statistical data which serves as the basis for comparison in order to establish sprint interval training as a viable, aerobic capacity increasing conditioning program. Although substrate utilization is generally different for aerobic states via the TCA cycle and anaerobic states via glycolysis, energy generation and muscle contraction is ultimately through dephosphorylation of ATP.

In anaerobic glycolysis, citrate serves to inhibit the activity of the glycolytic enzyme phosphofructokinase. In the mitochondrial, and thereby aerobic, TCA cycle, citrate is an intermediate formed from the condensation of oxaloacetate with acetyl-CoA catalyzed by the key component enzyme citrate synthase.[15] It is important to present citrate synthase as a component of an aerobic, mitochondrial process as it is a measurable quantity in sprint interval training research as a marker of muscle oxidative potential via muscle biopsies.

Generally defined, sprint interval training or high-intensity interval training is a form of exercise that alternates short bursts of high intensity exercise with slower, low intensity recovery periods within a single workout. However, in order to properly examine its role in improving aerobic capacity, a more structured definition must be established by presenting sprint interval training protocols as applied in several key studies and the data yielded. The following research protocols serve as the basis for developing a working definition of sprint interval training.

MacDougall et al's research conclusions were based on a clinical trial of three training sessions per week on alternate days for 7 total weeks.[5] Utilizing a Wingate Test, training sessions consisted of 30 second maximum-effort intervals against a constant force on a mechanically braked pan-loaded Monarch cycle ergometer. The Wingate Test is an anaerobic test consistent with sprint interval training characteristics which is typically employed to stress an individual's anaerobic capacity in order to measure both anaerobic power and capacity.[16] Week one training sessions consisted of four intervals with 4 minute recovery periods. Two intervals were added per week in weeks two through four while recovery periods remained at 4 minutes. The final three weeks consisted of 10 intervals with recovery periods decreasing by 30 seconds each week.

Twelve kinesiology graduate and undergraduate physically active males engaged in weight training, jogging, and intramural sports, but not varsity athletics, were selected to participate in sprint interval training as a basis for comparison to previous endurance training research[17,18] No additional exercise training during the study was permitted. Of the subjects meeting the criteria, the mean age was 22.7 ± 2 $years$, height 175 ± 6 cm, and body mass 73.4 ± 6.2 kg.

Sprint interval training effects on V_{O_2} max and enzymatic adaptations are presented in the following figures 2-4:

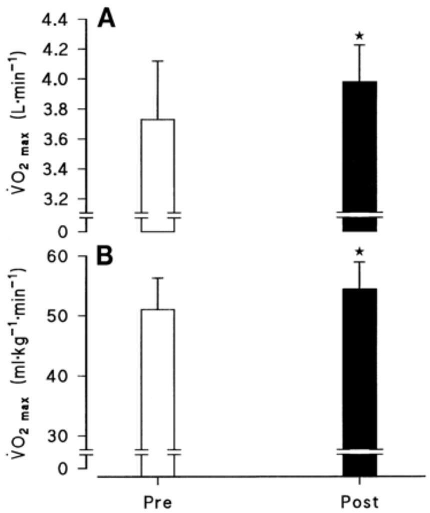


Fig. 2.
Maximum oxygen consumption (V˙o 2 max) before (Pre) and after (Post) training. Values are expressed in absolute units (l/min) in A and relative to body mass ($ml \cdot kg^{-1} \cdot min^{-1}$) in B. Values are means ± SD; n = 12 men.* $P < 0.05$.

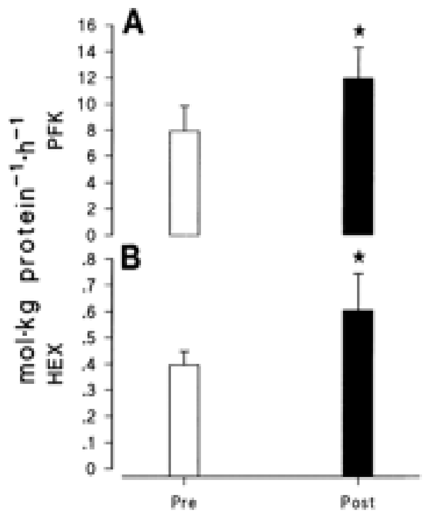


Fig. 3.
Maximal enzyme activity for phosphofructokinase (PFK;A) and hexokinase (Hex;B) before and after training. Values are means ± SD; n = 9 men.* $P < 0.05$.

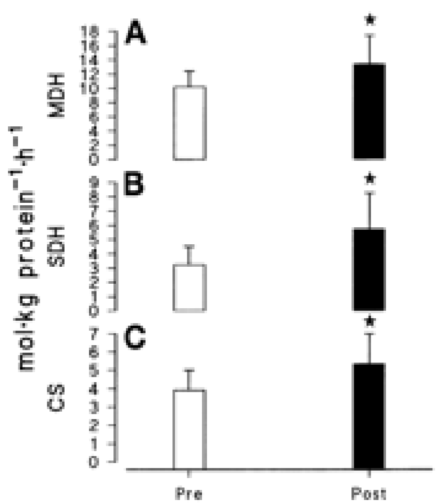


Fig. 4.
Maximal enzyme activity for malate dehydrogenase (MDH;A), succinate dehydrogenase (SDH;B), and citrate synthase (CS;C) before and after training. Values are means ± SD; n = 9 men.* $P < 0.05$.

Interpretation of the data presented shows maximum oxygen consumption (V_{O_2} max) increased from pre-training levels of 3.73 ± 0.13 to post-training levels of 4.01 ± 0.08 l/min in addition to V_{O_2} max relative to body mass increases from 51.0 ± 1.8 to 54.5 $ml/kg * min$ as illustrated in Fig. 2. Furthermore, Fig. 4 illustrates significant increases in oxidative marker enzymes with post-training malate dehydrogenase, succinate dehydrogenase, and citrate synthase activities increased ~29%, ~65%, and ~36% respectively. Those increases in oxidative marker enzymes are comparable to earlier research conducted by Linossier et al,[19] Cadefau et al,[20] and Roberts et al[21] which served as the impetus for MacDougall et al to examine the notion that sprint interval training improves muscle oxidative capacity.

Linossier et al conducted 7 weeks of repeated 5-second all out sprints in ten students exercising on a cycle ergometer after which muscle biopsies were taken from the vastus lateralis. Biopsies demonstrated increases in the slow twitch fibers with concomitant decreases in fast twitch fibers. Slow twitch fibers are characteristically utilized in endurance training as evidenced by the fact that athletes with the highest aerobic and endurance capacities, including distance runners and cross-country skiers, have the highest percentage of slow twitch fibers.[22]

As a result, Linossier et al concluded that adaptive reactions for slow twitch fibers exhibited greater oxidative capacity following high intensity intermittent training. Cadefau et al investigated the effects of 8 months of sprint interval training on three groups of young athletes, both male and female. Again biopsies of the vastus lateralis were evaluated. Evaluations demonstrated significant increases in succinate dehydrogenase, thereby demonstrating oxidative metabolic adaptations following an extended period of sprint interval training.

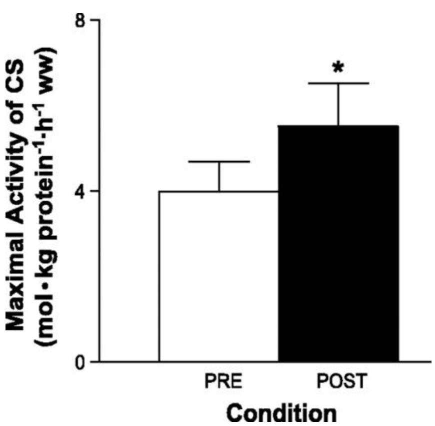
Roberts et al conducted 16 training sessions over 5 weeks on four male subjects. Sprint-type training consisted of 8 200-m runs at 90% maximum speed, separated by 2-minute rest periods. Post-training biopsies of the lateral head of the gastrocnemius were analyzed for enzymatic activity. Analysis identified increases in succinate dehydrogenase and malate dehydrogenase which were not considered relevant at the time because research was being conducted on the anaerobic adaptations to sprint interval training rather than the aerobic adaptations as addressed in the current research.

It is of particular interest with regard to sprint interval training that MacDougall et al noted that the "significant increase in V_{O_2} max and the large increases in muscle oxidative enzyme activity were somewhat *unexpected* given the nature of the training stimulus and its brevity." Typically, increases in V_{O_2} max and muscle oxidative enzyme activity are associated with several hours of aerobic, endurance training per week as demonstrated in research by Phillips et al[23] and Wibdom et al.[24] The former showed that maximal activity of succinate dehydrogenase, used to estimate muscle oxidative potential, was significantly elevated (+41%; $P < 0.01$) only after 31 days of prolonged endurance training consisting of 2 hours of submaximal cycling. As a result, conclusions were made that adaptations in muscle metabolism linked to increases in muscle mitochondrial capacity required not only endurance training, but training carried out over the course of an entire month. The latter research demonstrated mitochondrial ATP production rate (MAPR) and mitochondrial enzyme activities via increases of 40% and 45% in citrate synthase and glutamate dehydrogenase activity respectively after 6 weeks of endurance training. Citrate synthase catalyzes the initiation of the TCA cycle, and glutamate dehydrogenase catalyzes the formation of α-ketoglutarate, a TCA cycle component. In addition, mitochondrial enzyme activity increases were observed in cytochrome-c oxidase and succinate cytochrome-c reductase, two other enzymes used as markers to demonstrate increased aerobic capacity in sprint interval training.

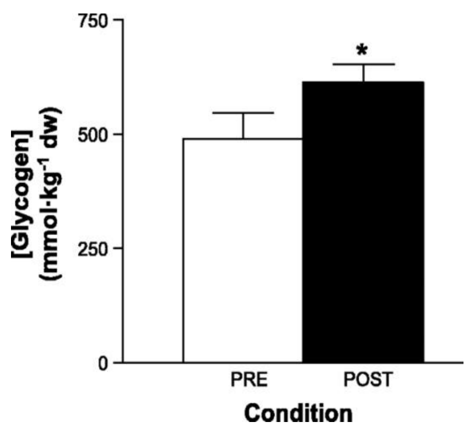
Burgomaster et al recognized the significant contributions of aerobic energy metabolism during repeated sprinting as presented in prior research.[6] One study examining that contribution tested eight male subjects performing two cycle ergometer sprints.[25] The first sprint lasted 30 seconds, and the second sprint lasted either 10 or 30 seconds with 4 minutes of rest between sprints. Two separate tests were conducted which demonstrated decreases in anaerobic ATP turnover, indicated primarily by a 45% decrease in glycolysis. However, the total decrease in work done on the second sprint was only 18%. That discrepancy between anaerobic energy release and power output during the second sprint was determined to be due to the increased contributions of aerobic metabolism.

Given the significant contributions of aerobic energy metabolism during repeated sprints,[25,26] and the increase in muscle oxidative potential indicated by changes in citrate synthase as a marker enzyme,[5,11,27] Burgomaster et al presented data from a clinical trial comprised of six training sessions spread over 14 days.[6] The eight subjects included in the research were recreationally active and 22 ± 1 year of age with V_{O_2} peak equal to 45 ± 3 $ml/kg * min$. Recreationally active was defined as individuals participating in some form of exercise including jogging, cycling, and aerobics two to three times per week, but not in any structured training program. Each session consisted of repeated 30 second “all-out” efforts on an electronically braked cycle ergometer against a resistance equivalent to 0.075 kg/kg $body\ mass$ with 4 minute complete rest or low cadence (< 50 rpm) recovery periods between tests. The number of intervals completed per training session increased from 4 to 7 over the first five training sessions and four intervals on the sixth and final session. This particular sprint interval protocol was designed to test the hypothesis that six sessions of sprint interval training performed over two weeks would increase citrate synthase activity, V_{O_2} peak, and the endurance time to fatigue during cycling.

Pre- and post-training biopsies of resting muscle were measured for citrate synthase which has been demonstrated to be the most used enzymatic marker of muscle oxidative potential, existing in constant proportions to other recognized mitochondrial enzymes.[5,11,27] Post-training maximal activity of citrate synthase and resting muscle metabolic concentrations of glycogen increased by 38% and 26% respectively as summarized below:



Maximal activity of citrate synthase (CS) measured in resting muscle biopsy samples obtained before and after a 2-wk sprint training protocol. Values are means ± SE for 8 subjects, ww, Wet wt. *$P < 0.05$.



Muscle glycogen concentration measured in resting biopsy samples obtained before and after a 2-wk sprint training protocol. Values are means ± SE for 8 subjects. dw; Dry weight. *$P < 0.05$

The post-training increase of glycogen in resting muscle is an indication that the anaerobic, glycolytic pathway is not being activated as substantially as would typically be expected. The post-training maximal activity increase in citrate synthase is comparable to other reported increases resulting from sprint interval training, with an emphasis that the other research utilized protocols with significantly more intervals.[5,11,12,27]

Other research which failed to show an increase in citrate synthase was discredited due to the fact that the protocols implemented very short sprints lasting less than 10 seconds. Ten second sprint intervals is in stark contrast to those studies reporting increases in citrate synthase which utilize 15-30 second maximal effort sprints. More importantly, in developing a case for oxidative, metabolic adaptations to sprint interval training, the post-training increase in citrate synthase is comparable to increases resulting from endurance training[13,14] However, the increase in citrate synthase is in contrast to still other research reporting no changes in muscle oxidative potential from short-term endurance training.[28]

In addition, post-training cycling endurance capacity improvements ranged from 81 to 169% compared with pre-training baseline. Also noted as the “primary novel findings” of the study were muscle oxidative potential increases with endurance time to fatigue at ~80% V_{O_2} peak cycling increasing 100%. From that data, it was concluded that short, repeated bouts of all out cycling (sprint interval training) “favorably altered the resting metabolic profile of human skeletal muscle.”

Subsequent to the aforementioned “primary novel findings,” Gibala et al directly compared sprint interval training and endurance training in a standardized manner in order to demonstrate the similar metabolic and performance adaptations that time- efficient sprint interval training could induce.[29] While it had been speculated that sprint interval training could induce such adaptations,[30] the research is considered to be the first that actually demonstrated that fact by recruiting sixteen physically active men participating in recreational exercise two to three times per week and randomly assigning eight to a sprint interval protocol and eight to an endurance protocol in order to make the direct comparison. Sprint interval training protocol was similar to that previously described.[6] Endurance training protocol consisted of 90-120 minutes of continuous cycling at 65% V_{O_2} peak with 1-2 days of recovery between training sessions. The endurance training protocol was patterned after previous work,[14,31] and the entire study was designed to produce an equal number of training sessions on the same days with the same number of recovery days between the two groups. For the basis of performance adaptations, performance tests included 50 kJ and 750 kJ, total energy expended, cycling time trials with the former taking approximately 2 minutes and the latter approximately 1 hour to complete. As a result, the energy contributions from oxidative and non-oxidative metabolism were considerably different. Based on findings from prior research,[5,6,13,14] it was hypothesized that both types of training would increase muscle oxidative capacity, and therefore, improve the 750 kJ time trial performance because of the energy contribution from aerobic metabolism required to complete the 1 hour task. In contrast, only the sprint interval training group was expected to improve on its 50 kJ time trial performance due to the increased muscle buffering capacity resulting from sprint interval training and the energy contribution from anaerobic, non-oxidative, metabolism required to complete a 2 minute task.

While completing time trials, subjects only received feedback based on the amount of work completed. Work completed was presented as “distance covered” on a computer monitor. In other words, 50 kJ and 750 kJ corresponded to 2 km and 30 km respectively, with subjects being provided units of distance rather than work completed during time trials.

The distinctive purpose of the study was to compare exercise capacity and molecular and cellular adaptations between the sprint interval group and endurance training group after two weeks of training. With regard to exercise capacity, i.e. the ability to sustain a given workload for a longer duration of time, thereby decreasing the time necessary to complete a task, or attain a higher average power output, measured in watts (W) over a predetermined distance or time,[32,33] both training protocols demonstrated comparable improvements in the 750 kJ cycling test. Recalling that the 750 kJ cycling test is characteristic of aerobic training, the time required to complete the cycling time trial post-training decreased by 10.1% in the sprint interval training and 7.5% in the endurance training groups. The mean power output also improved with an increase from 212 ± 17 to 234 ± 16 W and 199 ± 13 to 212 ±12 W in the sprint interval training and endurance training group respectively.

Changes in skeletal muscle substrate metabolism to identify molecular and cellular adaptations were determined via muscle biopsy samples obtained pre- and post-training. Biopsy samples from both training groups revealed similar increases in muscle oxidative capacity. Muscle oxidative capacity was indicated by an increase in the maximal activity of cytochrome c oxidase, a respiratory electron transport chain enzyme located in the mitochondrial membrane, and cytochrome c oxidase subunits II and IV protein content.

Additional findings included an increase in muscle buffering capacity and resting muscle glycogen. Muscle buffering capacity increased post-training by 7.6% for the sprint interval training group and 4.2% for the endurance training group. The resting muscle glycogen content showed post-training increases of 28% and 17% for the sprint interval training and endurance training groups respectively.

Following Gibala et al's research, Burgomaster et al conducted research to not only confirm but expand on their and Gibala et al's work described above which showed similar increases in muscle oxidative capacity between sprint interval and endurance training. Because the prior research findings were based on short, two week studies, it could be argued that the equivalent comparisons were the result of sprint interval training stimulating rapid skeletal muscle remodeling; whereas, the full effects of endurance training with slower adaptive processes could not be appreciated over the course of only two weeks.[7] Moreover, the single marker of muscle oxidative capacity, citrate synthase, in the former study provided a limited perspective on the potential for metabolic changes. Therefore, new research protocols were designed to "compare changes in markers of skeletal muscle carbohydrate and lipid metabolism and metabolic control during matched-work exercise, before and after 6 weeks of low-volume sprint interval training or high-volume endurance training."

For the purpose of this study, sprint interval training consisted of three training sessions per week for six weeks. Training sessions consisted of four, five, and six repeated Wingate Tests during weeks 1 & 2, weeks 3 & 4, and weeks 5 & 6 respectively. Recovery periods for all training sessions consisted of low cadence (< 50 rpm) cycling for 4.5 minutes. Endurance training, by contrast, consisted of continuous cycling at ~65% of V_{O_2} peak for 40 - 60 minutes, 5 days a week for 6 weeks. By design, the weekly exercise volume as defined by amount of energy expended, or work done, in applying a force of one newton through a distance of one meter, also known as a joule, was ~90% lower for subjects performing the sprint interval training protocol.

Prior to conducting the trial, it was hypothesized that despite significant training volume and time commitment differences, sprint interval training and traditional endurance training would induce comparable adaptations in muscle oxidative capacity and improve performance. In order to exemplify the significant contrast in training volume and time commitment, a summary of sprint interval training versus endurance training protocols is presented:

Summary of sprint interval training (SIT) and endurance training (ET) protocols

Variable	**SIT Group (n=10)**	**ET Group (n=10)**
Protocol	30 s x 4-6 repeats, 4.5 min rest (3 x per week)	40-60 min cycling (5 x per week)
Training Intensity	"All out" maximal effort	65% of V_{O_2} peak*
Workload	(~ 500 W)	(~ 150 W)
Weekly Training	~ 10 min	~ 4.5 h
Time commitment	(~ 1.5 h including rest)	
Weekly training volume	~ 225 kJ	~ 2250 kJ

*V_{O_2} peak, peak oxygen uptake

As noted from the presented data, weekly training volume and time commitment for sprint interval training was 90% (~225 kJ versus ~2250 kJ per week) and ~67% (~1.5 h versus ~4.5 h) lower than endurance training. Furthermore, with regard to time commitment, it is important to recognize that a significant majority of the training time is actually spent in recovery between short, intense all out bursts of cycling.

Similarly, with respect to previous research, adaptations in selected skeletal muscle CHO markers, lipid metabolism, and metabolic control during exercise compared favorably to those seen in endurance training. In addition, muscle oxidative capacity and citrate synthase maximal activity increases after 6 weeks of sprint interval training were similar to the increases in cytochrome c oxidase after 2 weeks training in prior studies. That comparison lead to the conclusion that most of the increases in mitochondrial content occur early in the training process. The premise that "pulmonary oxygen (O_2) uptake in the transition from rest to constant cycle exercise does not rise instantaneously though certainly very rapidly to a level corresponding to the amount of work performed"[7] was first presented by Krogh & Lindhard[34] and further supported by Krustrup[35] et al whose research demonstrated that at the initial phase of high intensity interval training, but not low intensity, oxygen uptake, blood flow, and vascular conductance all increased.

Other data presented to emphasize comparable improvements in exercise capacity between sprint interval training and endurance training included heart rate, respiratory exchange ratio, ventilation, and V_{O_2}. Improvement in heart rate is arguably one of the most recognizable physical adaptations to the average layman that represents increased aerobic capacity as summarized in the following work from Burgomaster et al which represents pre- and post-training cardiorespiratory data:[7]

Cardiorespiratory data during cycling exercise at 65% V_{O_2} *peak* before and after 6 weeks of sprint interval training (SIT) or 6 weeks of endurance training (ET)

	SIT Group		ET Group	
	Pre	**Post**	**Pre**	**Post**
Heart Rate ($beats/min$)	160 ± 5	151 ± 6	157 ±5	144 ± 5
RER*	0.977 ± 0.01	0.965 ± 0.01	0.967 ±0.01	0.941 ±0.01
Ventilation (l/min)	48 ± 3	42 ± 3	47 ± 5	42 ± 4
V_{O_2} ($ml/kg\ min$)	27.7 ± 1.0	26.6 ± 1.0	26.1 ±1.1	25.3 ±1.8

*RER, respiratory exchange ratio; V_{O_2}, oxygen uptake

With prior research having established sprint interval training effects on oxidative capacity and performance based primarily on Wingate Tests, Little et al presented conclusions from a clinical trial designed to determine the metabolic, molecular, and performance adaptations to a "more practical method of low-volume high intensity training."[8] Other research had demonstrated metabolic adaptations with more practical sprint interval training protocols,[36,37] but Little et al questioned the time efficiency of those protocols as training sessions exceeded 60 minutes in duration. It was believed that Wingate Tests, with maximal effort, variable-load exercise protocols were not practical due to the extreme demands of training and the specialized equipment required. Furthermore, safety concerns were considered since the extreme demands of training via Wingate Tests might not be well tolerated by individuals, i.e. the general public, wanting to receive the established benefits of sprint interval training.

Seven men, 21 ± 0.4 $years$ with V_{O_2} peak = 46 ± 2 $ml/kg * min$, completed the clinical trial consisting of six training sessions over 2 weeks. Each session consisted of repeated 60 second efforts of high-intensity cycling at ~100% peak power output. High-intensity cycling efforts corresponded to the peak power achieved (355 ± 10 W) at the end of the ramp V_{O_2} peak test. Each interval was separated by 75 seconds of low intensity (30 W) for recovery. Eight high-intensity intervals were completed in sessions one and two, ten intervals in sessions three and four, and twelve intervals in sessions five and six. Approximately 20-29 minutes total time committed per session including 3 minute warm-up at 30 W resulted in a 2 week total time commitment of 2 hours and 25 minutes.

As with prior research, performance adaptations were measured utilizing 50 kJ and 750 kJ cycling time trials.[29] Post-training tests documented time trial improvements of 11% and 9% with concomitant increases in mean power output from 397 ± 27 to 436 ± 22 W for the 50 kJ and 200 ± 7 to 221 ± 8 W for 750 kJ time trial. In addition, vastus lateralis biopsy analysis for mitochondrial enzyme activity monitored via cytochrome c oxidase and citrate synthase showed post-training increases of ~29% and ~16% respectively, further confirming the findings of earlier research.[5,11,27,29]

In an effort to elucidate already published research, Little et all further explored proposed regulators of mitochondrial biogenesis including peroxisome proliferator-activated receptor-γ coactivator (PGC)-1α, silent information regulator T1 (SIRT1), and mitochondrial transcription factor A (Tfam) in order to present potential mechanisms to adaptation. PGC-1α is a transcription coactivator regulating cellular metabolism which stimulates mitochondrial biogenesis and promotes muscle tissue remodeling to a metabolically more oxidative composititon.[38] SIRT1 is an NAD^+-dependent enzyme which deacetylases and activates PGC-1α.[39-41] Tfam is a mitochondrial transcription factor induced by PGC-1α which participates in mitochondrial genome replication.[42]

Evaluation of the aforementioned regulators of mitochondrial biogenesis showed significant post-training increases. While the total protein content of PGC-1α was unchanged, the nuclear abundance of PGC-1α increased by ~24%. Total SIRT1 content and Tfam total protein content had respective increases of ~56% and ~37%. Furthermore, muscle biopsy evaluation identified post-training increases of 17% in resting muscle glycogen and 119% in the insulin-stimulated glucose transport protein GLUT4. Since GLUT4 mediates glucose uptake into skeletal muscle to be stored as glycogen or oxidized for energy production,[43] other endurance training based research has linked post-training GLUT4 and resting muscle glycogen increases to improved insulin-sensitivity.[44-46]

Engaging in traditional aerobic and resistance exercise for several hours a week is the recommendation for improving glycemic control.[47,48] However, because research had demonstrated comparable improvements in aerobic capacity, function, and performance resulting from sprint interval training,[5-8,11,12,27-30,35-37] Babraj et al examined the capability of sprint interval training to improve insulin action and likewise glycemic control.[49] With lack of time being the number one perceived barrier to engaging in regular, i.e. traditional endurance training, exercise,[50,51] the research was conducted to propose sprint interval training as a time-efficient alternative to traditional endurance training for reducing metabolic disease risk factors focusing primarily on preventing and treating type 2 diabetes.

Sixteen young men (age: 21 ± 2 $years$; BMI: 23.7 ± 3.1 kg/m^2; V_{O_2} peak: 48 ± 9 $ml/kg * min$) completed a sprint interval training protocol comprised of 6 sessions over 2 weeks simulating that used by Burgomaster et al.[6] Pre- and post-training glucose, insulin, and nonesterified fatty acid responses to an oral glucose tolerance test (OGTT, administered by 75 gram oral glucose load) were determined in addition to endurance performance.

Endurance performance, as determined by comparable cycling time trials as previously described,[8,29] was evaluated in order to "provide an integrated physiological readout to facilitate comparison of the present study with previous studies which provided data from muscle biopsy samples." Evaluation of endurance performance determined a post-training increase of 6%, on par with prior research findings.

Post-training data analysis revealed no change in fasting plasma glucose concentrations. In contrast to the pre-training OGTT demonstrating a ~1.5 $mmol/l$ elevation in plasma glucose concentration 60 minutes after the 75-g glucose bolus, post-training OGTT plasma glucose concentration was not elevated after 60 minutes. (pre: 0 min: 5 ± 0.1, 60 min: 6.5 ± 0.4 vs post: 0 min: 5.0 ± 0.1, 60 min: 5.0 ± 0.2 $mmol/l$; P < 0.0001). Furthermore, post-training plasma glucose area under the curve (AUC) was significantly reduced (AUC pre: 664 ± 103 vs AUC post: 585 ± 65 $mmol * min/l$; P < 0.001, 12% reduction).

Fasting insulin glucose concentrations also demonstrated no change post-training. While pre- and post-training OGTT plasma insulin concentrations were elevated 60 minutes after the 75-g glucose bolus, the increase was significantly attenuated post-training (pre: 0 min: 10.5 ± 1.6, 60 min: 74.0 ± 8.9 mU/l; P < 0.05 vs post: 0 min: 10.6 ± 1.6, 60 min: 42.6 ± 6.0 mU/l; P < 0.01). Plasma insulin AUC was also significantly reduced post-training (AUC pre: 4226 ± 1912 vs AUC post: 2654 ± 1252 $mU * min/l$; P < 0.001; 37% reduction) while insulin sensitivity, as determined by calculating the Cederholm index,[52] was significantly improved (pre: 80 ± 6 vs post: 98 ± 5 $mg * l^2/mmol * mU * min$; P < 0.01).

Impaired suppression of plasma nonesterified fatty acids after glucose ingestion impairs insulin's ability to suppress hepatic glucose output and thereby contributes to glucose intolerance.[53] Consequently, a high plasma nonesterified fatty acid concentration is an identifiable risk marker for the deterioration of glucose tolerance.[54] With regard to Babraj et al's research, post-sprint interval training analysis identified a trend towards a decrease in baseline, i.e. fasting, plasma nonesterified fatty acid concentrations (pre: 350 ± 36 vs post: 290 ± 39 $\mu mol/l$; P = 0.058). Post-training OGTT plasma nonesterified fatty acid concentration was decreased to a much greater extent 60 minutes after the 75-g glucose bolus than pre-training (pre: 0 min: 350 ± 36, 60 min: 255 ± 48 $\mu mol/l$; P < 0.01 vs post: 0 min: 290 ± 39, 60 min: 153 ± 17 $\mu mol/l$; P < 0.001; pre 60 min: 255 ± 48 vs post 60 min: 153 ± 17 $\mu mol/l$; P < 0.05). As with plasma glucose and insulin AUC, plasma nonesterified fatty acid AUC was substantially reduced post-training (AUC pre 31584 ± 13205 vs AUC post 23370 ± 8630 $\mu mol * min/l$; P < 0.001, 26% reduction).

Conclusion

Engaging in regular physical activity has numerous health benefits[55-57] with research demonstrating that traditional endurance training induces metabolic adaptations which lead to improved muscle oxidative capacity, endurance performance, and correlate with improved insulin action and glycemic control.[44-48] As a result, engaging in regular endurance training exercise has long been the clinical strategy employed to modify the risk factors of metabolic diseases including type 2 diabetes and cardiovascular disease. However, current endurance training exercise program recommendations require a several hours per week time commitment. Given that lack of time is considered the greatest perceived barrier to engaging in regular aerobic exercise,[50,51] the result is low- or non-compliance with statistics showing that most of the U.S. adult population is not sufficiently active and 26% is not active at all.[58] Therefore, the effectiveness of current exercise guidelines as a meaningful preventative health strategy with positive economic benefit by reducing medial expenditures has been questioned with sprint interval training proposed as an alternate clinical strategy.

Sprint interval training, typically characterized as an anaerobic activity, demonstrates comparable metabolic adaptations to endurance training in improved muscle oxidative capacity, endurance performance, insulin action, and glycemic control. In contrast, however, research has demonstrated that the time commitment to achieve sprint interval training adaptations equates to ~20 minutes over a two-week period[5-8, 11,12,27-30,35-37,49] rather than the several hours per week required to achieve endurance training adaptations. Therefore, sprint interval training, supported by additional evidence that higher training intensities demonstrate greater improvements in muscle metabolic parameters and clinical outcomes,[59] has been presented as a time-efficient training protocol which may promote increased compliance.

Continued research into optimizing exercise strategies from a clinical perspective is warranted as the long term effects of sprint interval training on the body's cardiopulmonary and muscular systems is unknown. Furthermore, routine high-intensity training may have deleterious effects on joint health. However, appropriately administered as a clinical strategy, the potency of sprint interval training in stimulating metabolic adaptations and the potential for increased compliance may prove to be a more effective preventative health strategy than traditional endurance training in reducing the risk factors of metabolic disease.

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